

FURNACE DESIGN OF 210 MW CIRCULATING FLUIDIZED BED BOILER-NUMERICAL INVESTIGATION

The original paper [↗](#) contains 16 sections, with 10 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Introduction

Guangwen Xu et al. [23] proposed equations for local voidage at the wall and centre of the furnace only. According to Guangwen Xu et al [23], the wall voidage can be considered as $\hat{\mu}_{mf}$ when $\hat{\mu}_{avg}$ is less than 0.75.

SUMMARY PASSAGE 2

Design Of Furnace

Along with dimensions, A furnace design also provides the value of water wall surface area. The arrangement of components in the furnace is as shown in figure 1 and the following steps are needed for the design of furnace of CFB boiler from P.Basu [4]. editor@iaeme.com

SUMMARY PASSAGE 3

Heat Duty Calculations

The boiler system of 210MW CFB boiler contains economizer, evaporator (water wall and wing wall), two superheaters (furnace superheater and backpass superheater) and one reheater. The heat duty of each boiler element is determined by steam parameters shown in Table 2. The operating steam parameters are assumed from the 210MW PC fired power plant and taken from the open literature.

SUMMARY PASSAGE 4

Heat Balance

The heat balance provides the information about furnace heat losses. They are heat carried by moisture and dry flue gases, calcinations loss, unburned carbon loss, ash, convection & radiation, sulfation credit and FD fan credit [3]. The boiler efficiency in % can be calculated by heat loss method which is given as

SUMMARY PASSAGE 5

Furnace Cross Section

Furnace cross section is a function of total flue gases developed in the combustor, density of gases and fluidization velocity. It is given by the following formula from Prabir Basu [3].

SUMMARY PASSAGE 6

Furnace Height

P.Basu and P.K Nag [14] suggested to the boiler designers that the height of the water wall can be considered for the height of furnace in the initial stage of design. The water wall height is given by the following expression from P.Basu [4].

SUMMARY PASSAGE 7

Voidage And Pressure Drop Profile

$a(U - U_{mf}) = 3.5 \times 1670 d$ (15) In order to evaluate the voidage profile, mean voidage of dense region ' μ_d ' and furnace exit voidage ' μ_e ' are needed. J.Adanez et al [6] was used an expression for calculating the voidage at the exit of furnace. It is given by

SUMMARY PASSAGE 8

Figure.3: Heat Losses In Cfb Boiler

The depth of lower furnace was taken at 40% of primary air. The total dimensions of CFB furnace for 210MW plant operating with Gujarat lignite are shown in figure 6. Figure 7 shows the theoretical investigation of axial voidage distribution from the point of injection of secondary air to the riser exit by using equation (14). This profile was effected mainly by solid circulation rate, superficial gas velocity and particle diameter.

SUMMARY PASSAGE 9

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The furnace design of CFB boiler was investigated numerically by using operating steam parameters of 210MW steam power plant and analysis of Gujarat lignite and limestone. The cross sectional area of upper furnace was calculated as 137m² at superficial gas velocity. The furnace height was measured as 30m by using projected area heat transfer coefficient of water wall and mean bed voidage at minimum solids circulation rate.

SUMMARY PASSAGE 10

Abed

Depth of furnace (m) Ca/S Calcium to sulfur molar ratio CD Drag coefficient dp